

A Practical Guide to Ridge regression

TOPIC -- RIDGE REGRESSION

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

-- 01 --

$$\hat{x} = (A^T A)^{-1} A^T b = (A^T A + \Gamma^T \Gamma)^{-1} A^T b. \quad \{\displaystyle \hat{\mathbf{x}} = (\mathcal{A})^{\mathsf{T}} \mathcal{A})^{-1} \mathcal{A}^{\mathsf{T}} \mathbf{b} = (A^{\mathsf{T}} A + \Gamma^{\mathsf{T}} \Gamma)^{-1} A^{\mathsf{T}} \mathbf{b} .\}$$

-- 02 --

where the Wiener weights are $f_i = \frac{\sigma_i^2}{\sigma_i^2 + \alpha^2}$ $\{\displaystyle f_i = \frac{\sigma_i^2}{\sigma_i^2 + \alpha^2}\}$ and q $\{\displaystyle q\}$ is the rank of A $\{\displaystyle A\}$.

-- 03 --

Tikhonov regularization for linear equations Suppose that for a known real matrix A $\{\displaystyle A\}$ and vector $\mathbf{b}$ $\{\displaystyle \mathbf{b}\}$, we wish to find a vector $\mathbf{x}$ $\{\displaystyle \mathbf{x}\}$ such that

-- 04 --

$$\lim_{\lambda \downarrow 0} \hat{\beta}_\lambda = X^+ Y = \beta^0, \quad \{\displaystyle \lim_{\lambda \downarrow 0} \hat{\boldsymbol{\beta}}_\lambda = \mathbf{X}^+ \mathbf{Y} = \hat{\boldsymbol{\beta}}_0,\}$$

with X^+ $\{\displaystyle \mathbf{X}^+\}$ denoting the pseudoinverse of X $\{\displaystyle \mathbf{X}\}$.

-- 05 --

where we have used $\|\mathbf{x}\|_Q^2$ $\{\displaystyle \|\mathbf{x}\|_Q^2\}$ to stand for the weighted norm squared $\mathbf{x}^T Q \mathbf{x}$ $\{\displaystyle \mathbf{x}^T Q \mathbf{x}\}$ (compare with the Mahalanobis distance). In the Bayesian

interpretation P is the inverse covariance matrix of $\mathbf{b}$, $\mathbf{x}_0$ is the expected value of $\mathbf{x}$, and Q is the inverse covariance matrix of $\mathbf{x}$. The Tikhonov matrix is not explicitly included because the corresponding regularization term $\|\Gamma \mathbf{x} - \mathbf{x}_0\|_Q^2$ reduces to above with $\Gamma \mathbf{x}_0 = \mathbf{x}_0'$ and $Q = \Gamma^T Q' \Gamma$. For normal regularization where $Q' = I$, the Tikhonov matrix then appears in the Cholesky factorization $Q = \Gamma^T \Gamma$ and is considered a whitening filter. This generalized problem has an optimal solution $\hat{\mathbf{x}}$ which can be written explicitly using the formula

-- 06 --

and is zero elsewhere. This demonstrates the effect of the Tikhonov parameter on the condition number of the regularized problem. For the generalized case, a similar representation can be derived using a generalized singular-value decomposition. Finally, it is related to the Wiener filter:

-- 07 --

Generalized cross-validation estimator A common data-driven choice for λ is the minimizer of the cross-validation loss or its generalizations. For example, Grace Wahba proved that the optimal parameter, in the sense of generalized cross-validation minimizes

-- 08 --

In the context of arbitrary likelihood fits, this is valid, as long as the quadratic approximation of the likelihood function is valid. This means that, as long as the perturbation from the unregularised result is small, one can regularise any result that is presented as a best fit point with a covariance matrix. No detailed knowledge of the underlying likelihood function is needed.

-- 09 --

where $\|\cdot\|_2$ is the Euclidean norm. In order to give preference to a particular solution with desirable properties, a regularization term can be included in this minimization:

-- 10 --

Replacing the unknown β, ς^2 in the formula for λ^* with the corresponding $\hat{\beta}, \hat{\varsigma}^2$ thus gives the so-called plug-in estimator $\hat{\lambda}^*$ for the optimal λ^* .

-- 11 --

$$\hat{\beta} = (X^T X + \hat{\varsigma}^2 I)^{-1} X^T Y, \quad \hat{\varsigma}^2 = \frac{Y^T Y - \hat{\beta}^T X^T Y}{n - p}.$$

-- 12 --

where x and b may be of different sizes and A may even be non-square. The standard approach is ordinary least squares linear regression. However, if no x satisfies the equation or more than one x does—that is, the solution is not unique—the problem is said to be ill posed. In such cases, ordinary least squares estimation leads to an overdetermined, or more often an underdetermined system of equations. Most real-world phenomena have the effect of low-pass filters in the forward direction where A maps x to b . Therefore, in solving the inverse-problem, the inverse mapping operates as a high-pass filter that has the undesirable tendency of amplifying noise (eigenvalues / singular values are largest in the reverse mapping where they were smallest in the forward mapping). In addition, ordinary least squares implicitly nullifies every element of the reconstructed version of x that is in the null-space of A , rather than allowing for a model to be used as a prior for x . Ordinary least squares seeks to minimize the sum of squared residuals, which can be compactly written as

-- 13 --

the problem of a near-singular moment matrix $X^T X$ is alleviated by adding positive elements to the diagonals, thereby decreasing its condition number. Compared to the ordinary least squares estimator, the simple ridge estimator has an extra term λI in the denominator:

-- 14 --

where $\mathbf{Y}$ is the regressand or response vector, $\mathbf{X}$ is the design matrix, $\mathbf{I}$ is the identity matrix, and the ridge (or Tikhonov) regularization parameter $\lambda \geq 0$ serves as the constant shifting the diagonals of the moment matrix. It can be shown that this estimator is the solution to the least squares problem subject to the constraint $\mathbf{\beta}^T \mathbf{\beta} = c$, which can be expressed as a Lagrangian minimization: